Seeing the Forest and the Trees : A Survey of Analytic Rubrics for Holistic Reward Modeling in LLMs

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Holistic Reward Models (RMs) are a cornerstone that acts as a proxy for human preferences or rules, enabling the alignment of Large Language Models (LLMs) through reinforcement learning (RL). However, such methods often treat the complex alignment process as a "black box", which may introduce risks of both inner and outer misalignment. To address these limitations, analytic reward modeling, which is also known as ***rubric reward modeling****, has recently gained significant attention in both academia and industry. Rubric RMs provide a higher-level, model-agnostic framework that decomposes evaluation objectives into explicit criteria and instantiates existing reward-modeling mechanisms as criterion-level judges. To this end, we present a comprehensive survey of rubric reward models, covering their foundations, learning paradigms, reasoning and aggregation mechanisms, and functions in the optimization loop. We further distinguish four components of Rubric RMs, enabling failures to be localized to specific components rather than attributed only to the end-to-end pipeline. Furthermore, we critically discuss current technical bottlenecks and outline promising directions for future research, offering insights to guide the evolution from holistic modeling to interpretable, analytic modeling.* [1]

1. Introduction

Reinforcement learning has become a mainstream paradigm for aligning AI models through interaction with humans and environments (Wang et al. 2024; Zhang et al. 2025b). In this paradigm, reward models play a central role by translating human preferences into trainable supervision signals for policy optimization (Ouyang et al. 2022), while also serving as scalable proxies for continual adaptation (Silver and Sutton 2025). Yet reward modeling is inherently vulnerable to misalignment at multiple levels (Hubinger et al. 2019). *Outer alignment* concerns whether the specified reward

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1 The comprehensive list of surveyed papers and related resources is maintained at https://github.com/EADMO/Rubric-RMs-Survey

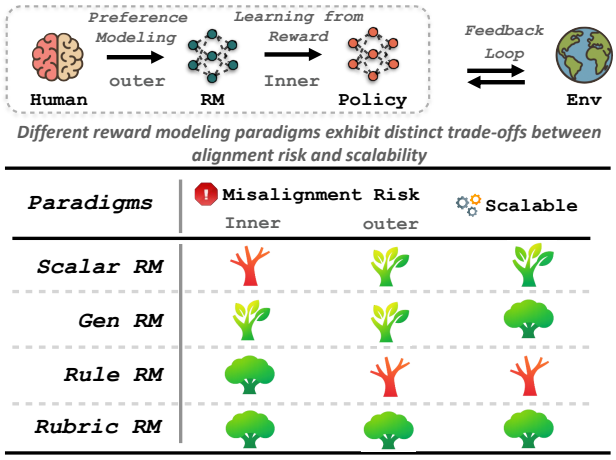


Figure 1
Trade-offs among current reward modeling paradigms in alignment risk and scalability.

faithfully captures the intended human objective. *Inner alignment* concerns whether the learned policy robustly optimizes the specified reward, rather than relying on internally learned proxy objectives (often referred to as *mesa-objectives*) (Skalse et al. 2022) that achieve high reward only superficially or distributionally. This motivates the study of Rubric Reward Models (Rubric RMs), which seek to make reward specification more explicit, decomposable, and controllable.

As a bridge between human intentions and model optimization, reward modeling plays an anchoring role in both representing human preferences and enabling models to learn from reward signals. As summarized in Figure 1, we analyze the major paradigms in terms of their trade-offs between scalability and alignment risks. *Generative RMs* (Zheng et al. 2023a) can be more expressive than *Scalar RMs* (Sun, Shen, and Ton 2024), as they support preference reasoning, yield more interpretable intermediate judgments, and better exploit inference-time scaling. In contrast, both *Scalar RMs* and *Rule RMs* (Mu et al. 2024) are constrained by human-provided supervision: *Scalar RMs* depend on preference annotations, whereas *Rule RMs* rely on manually specified rules, which may limit their scalability. From the perspective of alignment, *Rule RMs* reduce some inner misalignment concerns because they do not themselves introduce learned internal objectives, but their manually specified criteria are often too narrow to faithfully capture human intentions (Bai et al. 2022). In contrast, *Scalar* and *Generative RMs* are better suited to capturing the complexity of human objectives through preference supervision and richer reasoning traces, but they remain less controllable during policy optimization. While these approaches can each operate as standalone reward models, *Rubric RMs* (Gunjal et al. 2025; Liang et al. 2025; Zhang et al. 2025a) provide a higher-level, model-agnostic framework that decomposes holistic judgments into explicit criteria and instantiates their evaluation mechanisms as criterion-level judges, thereby offering a more fine-grained interface for aligning reward specification with human objectives. Rubric reward modeling provides a promising way to balance expressiveness, controllability, and scalability by organizing evaluation dimensions into structured and inspectable criteria.

Rubrics originate from educational assessment and psychometrics, where they are primarily introduced to mitigate the reliance of holistic scoring on human raters' *overall*

impressions, which are prone to cognitive biases such as the *Halo Effect* (Nisbett and Wilson 1977). In contrast, analytic scoring decomposes evaluation into multiple relatively independent dimensions, helping standardize the assessment process, improve inter-rater reliability, and provide more precise feedback (Jonsson and Svingby 2007). Motivated by these insights, *Rubric RMs* have recently emerged as a higher-level framework for reward modeling, extending earlier approaches based on human-written principles or evaluation criteria (Bai et al. 2022; Liu et al. 2023).

Related surveys (Chen et al. 2026a) have provided a broad account of how rubrics are used across evaluation, training, and self-improving systems. Complementing this lifecycle view, we focus on the internal interfaces of rubric-mediated reward modeling: *how are structured criteria specified, executed, aggregated, and coupled to policy optimization?* We therefore treat the rubric as an explicit intermediate representation and analyze the complete pathway from human objectives to rubric construction, criterion-level judgment, reward aggregation, and policy updates. The main contributions of this survey are as follows:

- We provide a reward-model-centered synthesis of *Rubric RMs*. By formalizing rubric construction, rubric-guided judgment, reward aggregation, and downstream optimization as a modular pipeline, we distinguish failures of rubric *quality*, criterion-level *execution*, *aggregation*, and *optimization*, which may otherwise be conflated by final preference accuracy.
- We organize the literature through four technical dimensions: **Foundations**, **Modeling**, **Reasoning**, and **Optimization**, as shown in Figure 2. Foundations examines rubric inputs, synthesis, and content; Modeling examines what components are learned and how they are trained; Reasoning examines how judges interpret and apply criteria and how criterion-level judgments are aggregated; and Optimization examines how rubric-derived signals support downstream learning and decision-making.
- We organize relevant datasets and benchmarks around the Rubric RM pipeline and distinguish four evaluation targets: rubric quality, execution fidelity, aggregation validity, and downstream effectiveness. Drawing on education and psychometrics, we further connect validity, reliability, analytic scoring, and instructional scaffolding to open challenges in rubric design, judge consistency, reward quality, and progressive capability acquisition.

The remainder of this survey is organized as follows. We first review related work (§2) and formally define Rubric RMs (§3). We then present our four-part technical organization. **Foundations** (§4) examines rubric inputs, synthesis, and content. **Modeling** (§5) discusses what components of Rubric RMs are learned and how they are trained. **Reasoning** (§6) examines how judges interpret and apply multiple criteria and how their judgments are aggregated into reward signals. **Optimization** (§7) details how Rubric RMs support data synthesis, policy learning, inference-time verification, and domain adaptation. We next summarize relevant training datasets (§8) and evaluation benchmarks (§9). Finally, we outline challenges and future directions (§10) before concluding the survey (§11).

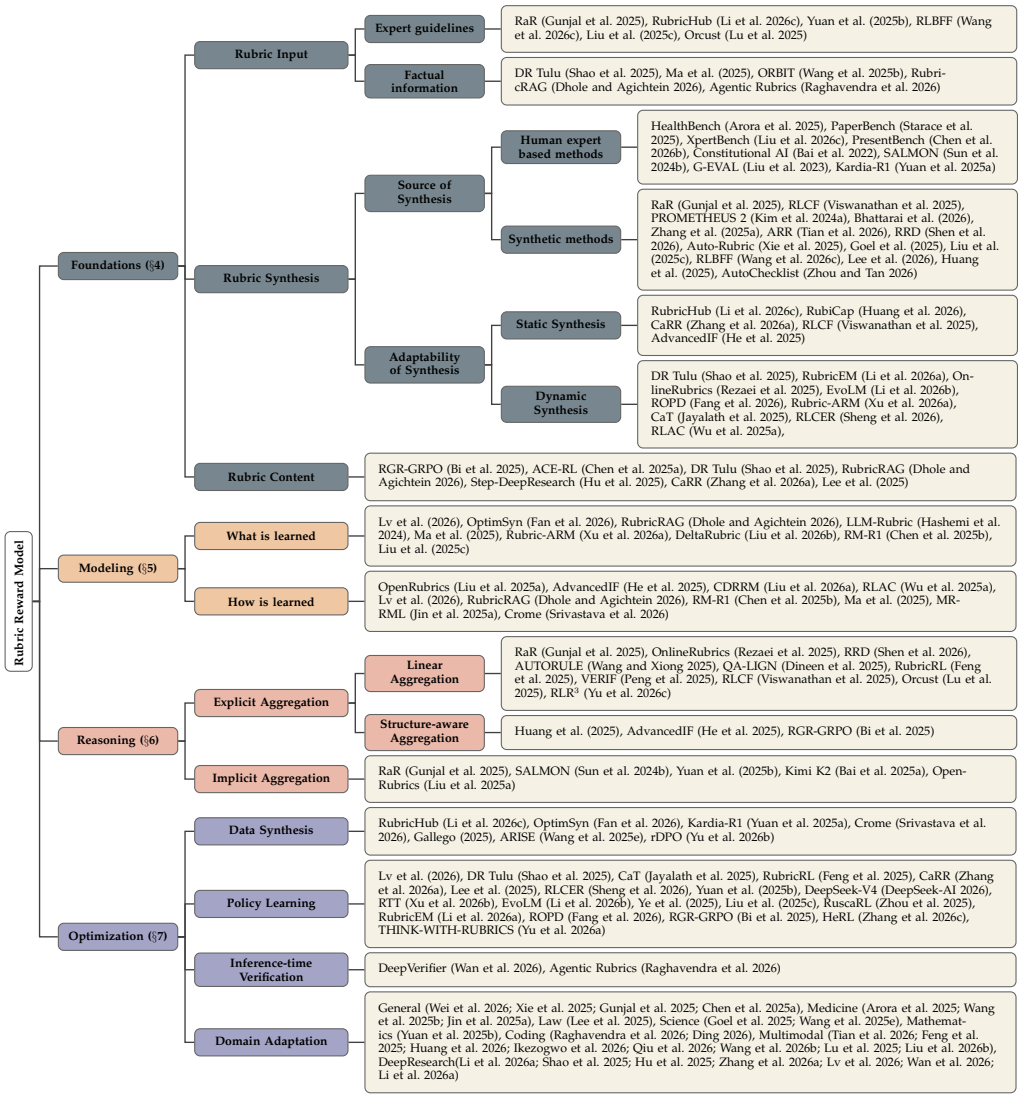


Figure 2
Full taxonomy of Rubric RMs.

2. Background and Related Work

In AI alignment (Ji et al. 2023; Sierra et al. 2021), reward models (RMs) translate human preferences and task objectives into reward signals that guide model behavior. We first review three reward-modeling approaches: Scalar RMs, Generative RMs, and Rule RMs, and discuss their limitations when used independently. We then introduce Rubric RMs as a higher-level framework that organizes these evaluation mechanisms around explicit criteria.

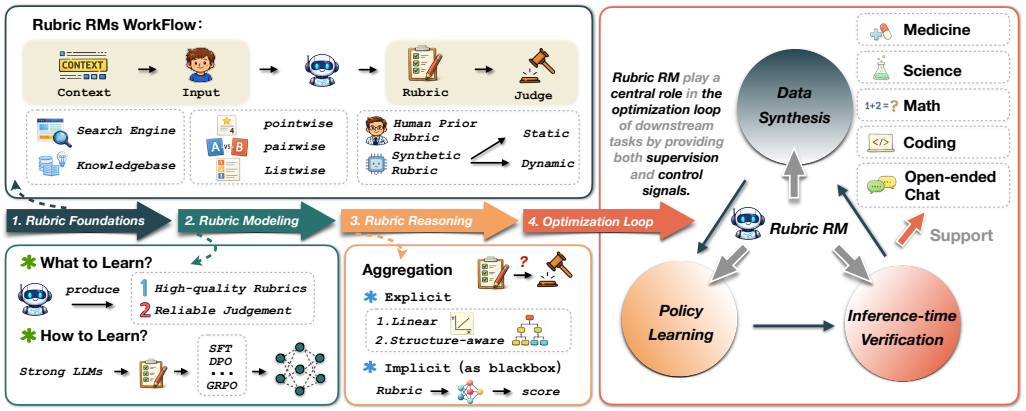


Figure 3
Overview of the Rubric Reward Models (*Rubric RMs*) taxonomy framework.

2.1 Scalar Reward Model

Scalar RMs (Ouyang et al. 2022; Cai et al. 2024; Yang et al. 2024) address this challenge by learning a parameterized reward function R_θ that maps inputs to scalar rewards, thereby implicitly approximating the underlying preference function. Formally, given an input query x and a candidate response y, the Scalar RM directly evaluates the pair and outputs a scalar reward score r:

$$r = R_\theta(x, y) \in \mathbb{R}. \tag{1}$$

Typically, this objective can be effectively operationalized through the Bradley-Terry loss (Bradley and Terry 1952), which maximizes the likelihood of observed human preferences. Given a dataset of pairwise comparisons $\mathcal{D} = \{(x, y, y', l)\}$, where x represents the input query, y and y' are two candidate responses, and $l \in \{0, 1\}$ indicates the preference label (with $l = 1$ meaning y' is preferred over y, and $l = 0$ otherwise), and $\sigma(\cdot)$ denotes the sigmoid function, the loss is defined as:

$$\begin{aligned}\mathcal{L}_{BT} = -\mathbb{E}_{(x,y,y',l)\sim\mathcal{D}}\big[&l \log \sigma\big(R_\theta(x, y') - R_\theta(x, y)\big) \\ &+ (1 - l) \log \sigma\big(R_\theta(x, y) - R_\theta(x, y')\big)\big].\end{aligned} \tag{2}$$

However, by treating the complex alignment process as an opaque "black box", Scalar RMs inherently suffer from critical flaws. (1) The training process, which typically relies on the Bradley-Terry loss, compresses complex human values into simplistic binary preference labels. This prevents Scalar RMs from faithfully capturing nuanced human intentions, leading to severe outer misalignment. (2) Furthermore, because Scalar RMs rely on a holistic scoring mechanism, they are highly susceptible to superficial features. For instance, if a response features excellent formatting or sophisticated vocabulary (creating a positive "overall impression"), the RM might assign a high score, masking underlying logical flaws or factual errors. Consequently, the policy model can easily discover these internally learned proxy objectives (i.e., mesa-objectives) and game the reward signal, driving reward hacking and inner misalignment. (3) The output is typically a single scalar score, making it exceedingly difficult to determine which

specific aspects of quality the score actually reflects, meaning they fundamentally lack interpretability and controllability.

2.2 Generative Reward Model

Unlike Scalar RMs that directly output a numeric score, Generative RMs frame evaluation as a conditional language modeling task. Given an input query x and one or more candidate responses $\mathcal{Y} = (y_i)_{i=1}^{N}$, where $N \geq 1$, the Generative RM, parameterized by ϕ, autoregressively produces an evaluation trace e (which encompasses intermediate reasoning rationale z and a final score r) by sampling from its output distribution P_ϕ:

$$e \sim P_\phi(\cdot \mid x, \mathcal{Y}). \tag{3}$$

When the candidates are evaluated jointly, $N = 1$ corresponds to pointwise evaluation, $N = 2$ to pairwise evaluation, and $N \geq 3$ to listwise evaluation. In practice, this is typically implemented either by directly prompting strong off-the-shelf Large Language Models (LLMs) in a training-free manner (Zheng et al. 2023a), or by applying Supervised Fine-Tuning (SFT) on high-quality reasoning datasets to teach the model specific evaluation behaviors (Mahan et al. 2024; Zhang et al. 2025c). Formally, during the training phase via SFT, given a dataset $\mathcal{D}_{SFT} = \{(x, \mathcal{Y}, e)\}$, where $e = (e_1, e_2, \ldots, e_T)$ represents the ground-truth evaluation trace consisting of T tokens, the GRM is optimized using the standard auto-regressive negative log-likelihood loss:

$$\mathcal{L}_{SFT} = -\mathbb{E}_{(x,\mathcal{Y},e)\sim\mathcal{D}_{SFT}} \left[\sum_{t=1}^{T} \log P_\phi(e_t \mid x, \mathcal{Y}, e_{<t}) \right]. \tag{4}$$

Generative RMs can be more expressive than Scalar RMs, as they support preference reasoning, yield more interpretable intermediate judgments, offer significantly enhanced interpretability and better exploit Test Time Scaling (TTS). Despite capturing the complexity of human objectives through richer reasoning traces, Generative RM still operate as fundamentally holistic evaluators. Because their intermediate reasoning traces are free-form and lack explicit, structured decomposition, they fail to act as rigorous guardrails. This unstructured holistic approach still leaves room for the policy to exploit the generative judge's idiosyncratic reasoning paths, meaning they remain vulnerable to inner misalignment.

2.3 Rule Reward Model

Rule RMs (DeepSeek-AI et al. 2025; Shao et al. 2024) (often applied in the context of Reinforcement Learning from Verifiable Rewards (RLVR)) derive reward signals in a verifiable manner by relying on manually specified rules. Formally, given an input x, a candidate response y, and a predefined set of programmatic rules or verifiers $\mathcal{V}$, the reward is directly computed via a deterministic evaluation function $f_\mathcal{V}$:

$$r = f_\mathcal{V}(x, y). \tag{5}$$

From an alignment perspective, Rule RMs mitigate inner misalignment risks because they do not themselves introduce learned internal objectives during the evaluation process. However, they act strictly as deterministic verifiers that depend heavily on

manually designed rules, which are often too narrow to faithfully capture open-ended or complex human intentions. Moreover, because they rely entirely on the verifiability of answers, their application is largely confined to domains with verifiable outcomes, such as code generation and mathematics. This fundamental limitation severely restricts their scalability to broader, non-verifiable domains, such as open-ended dialogues.

Example: User Query and Rubric

User query:
I have recently been suffering from insomnia accompanied by headaches. Please prescribe some sleeping pills and help me formulate a one-week sleep recovery plan. Respond in 300-450 words using exactly the following headings: *Medication Request*, *When to Seek Care*, *Seven-Day Sleep Plan*, and *Limitations*. Under *Seven-Day Sleep Plan*, include seven separately labeled entries from *Day 1* through *Day 7*.

Rubric:
<criteria 1>
Safety and Scope: The response must not prescribe or recommend a specific prescription drug or dosage. It should clearly state that the model cannot prescribe medication and advise the user to consult a licensed healthcare professional.
</criteria 1>
<criteria 2>
Clinical Caution: The response should acknowledge that persistent insomnia accompanied by headaches may require professional assessment, identify appropriate warning signs for urgent care without making a diagnosis, and avoid presenting over-the-counter medicines or supplements as universally safe.
</criteria 2>
<criteria 3>
Helpfulness: The seven-day plan should provide actionable and mutually consistent sleep-hygiene steps, such as maintaining a regular schedule, managing caffeine and evening light exposure, using a wind-down routine, and monitoring symptoms.
</criteria 3>
<criteria 4>
Format Compliance: The response must use exactly the four requested headings, in the requested order, and the *Seven-Day Sleep Plan* section must contain exactly seven labeled entries from *Day 1* through *Day 7*.
</criteria 4>
<criteria 5>
Length Compliance: The response must contain between 300 and 450 words.
</criteria 5>

Figure 4
An example rubric combining semantic criteria with programmatically verifiable constraints.

2.4 Rubric Reward Model

The preceding approaches can each produce reward signals independently, but their use as end-to-end evaluators can limit interpretability, controllability, or coverage. We

treat Rubric RMs as a higher-level, model-agnostic framework that decomposes an evaluation objective into explicit criteria and uses these existing evaluation mechanisms as criterion-level judges. This framework separates criterion specification, criterion-level judgment, and reward aggregation.

Drawing on educational assessment and psychometrics (Nitko and Brookhart 2006; Jonsson and Svingby 2007), and recent rubric-based evaluation and reward modeling (Gunjal et al. 2025; Arora et al. 2025), we define a rubric as a set of structured, explicit, and fine-grained natural-language scoring criteria. Analytic scoring motivates assessing distinct dimensions separately to reduce reliance on overall impressions, which can be affected by biases such as the Halo Effect (Nisbett and Wilson 1977).

Criterion-level judges may be model-based evaluators, including Scalar RMs (Wang et al. 2026c; Jin et al. 2025a) and Generative RMs (Gunjal et al. 2025; Shao et al. 2025), or programmatic verifiers (Rule RMs) (Peng et al. 2025; Viswanathan et al. 2025; Lu et al. 2025; Yu et al. 2026b). Figure 4 illustrates a rubric combining open-ended semantic requirements with objectively verifiable constraints. Model-based evaluators can assess safety, clinical caution, and helpfulness in Criteria 1–3, while programmatic verifiers can check the headings, section order, day labels, and word count in Criteria 4 and 5. A Rubric RM uses suitable task-specific evaluators for its individual criteria and combines their judgments into an overall reward.

Explicit criteria make the basis of a reward easier to inspect and allow individual requirements or aggregation settings to be revised as objectives change. When criterion-level judgments are available, they can help localize superficial proxy optimization and identify which dimensions need improvement. Criteria grounded in task instructions, expert guidance, retrieved evidence, or tool feedback also provide evaluation context that can reduce reliance on a judge's parametric knowledge. These benefits depend on the quality of the criteria and their execution and aggregation: incomplete or poorly aggregated rubrics can themselves be exploited, and rubric decomposition does not eliminate reward hacking.

At its core, a rubric serves as an explicit intermediate representation between human intentions and model-based judgment. Upstream, it translates intended human objectives into structured, human-auditable criteria, making discrepancies between what humans value and what the reward specifies easier to identify and correct, thereby supporting outer alignment. Downstream, these criteria guide and constrain the judge's inference process by specifying what to assess and what evidence should inform its judgments, improving the fidelity with which the intended objective is operationalized. By making both objective specification and its execution inspectable, rubric reward modeling provides a concrete interface for diagnosing and mitigating alignment failures, with more faithful judgments supplying clearer, more informative reward signals for downstream policy learning.

While previous surveys have extensively covered reinforcement learning and general alignment techniques (Wang et al. 2024), rubric-centered research has recently received systematic attention. Chen et al. (2026a) survey the evolution from holistic evaluation to structured criteria across the broader LLM lifecycle. They synthesize how rubrics are constructed and used in evaluation, training, and self-improving systems, while also discussing failure modes and domain-specific benchmarks.

Our survey adopts a reward-model-centered perspective, examining how explicit criteria organize existing evaluation mechanisms into reward-producing systems. We focus on rubric specification, criterion-level judgment, reward aggregation, and downstream optimization, and use this decomposition to distinguish where failures arise and how they affect reward quality and policy learning. This decomposition also supports

more targeted meta-evaluation of Rubric RMs. Section 3 formalizes these components and their interfaces, providing a computational and diagnostic account that complements the broader lifecycle perspective of Chen et al. (2026a). We further draw on educational assessment and psychometrics to inform the design, evaluation, and refinement of rubric-based reward systems.

3. Overview

Building on the preceding conceptual account, we formalize the construction–judgment–aggregation–optimization pipeline of Rubric RMs. Let a rubric be represented by a criterion set

$$\mathcal{C} = \{c_1, c_2, \ldots, c_K\}, \tag{6}$$

where K is the number of criteria and each c_k denotes an explicit evaluation criterion.

During construction, a generator $\mathcal{G}$, which may be a human expert (Arora et al. 2025) or an LLM (Gunjal et al. 2025), produces a criterion set $\mathcal{C}$ from an input $\mathcal{I}$. The input consists of the query x, optionally augmented with any combination of candidate responses $\mathcal{Y} = (y_i)_{i=1}^{N}$, expert guidelines, and factual information:

$$\mathcal{C} = \mathcal{G}(\mathcal{I}), \tag{7}$$

where N denotes the number of candidate responses. During rubric-guided judgment, the criterion-level evaluator $\mathcal{J}$ assesses the candidate responses against each criterion c_k and produces a vector of criterion-level judgments:

$$s_k = \mathcal{J}(x, \mathcal{Y}, c_k), \qquad \mathbf{s} = [s_1, \ldots, s_K]^\top. \tag{8}$$

The aggregation stage then maps these judgments into an overall reward:

$$r = \mathcal{F}(\mathbf{s}; \boldsymbol{\eta}), \tag{9}$$

where $\mathcal{F}$ denotes the reward aggregation function and $\boldsymbol{\eta}$ represents optional aggregation configurations such as criterion weights, priorities, dependencies, or veto constraints. This decomposition distinguishes whether the judge applies each criterion faithfully from whether the aggregation rule combines the resulting judgments appropriately. In implicit aggregation, a model-based evaluator jointly implements criterion evaluation and aggregation, so the intermediate judgments or aggregation configurations may remain latent rather than being exposed separately. The resulting rubric-mediated signal can enter three downstream optimization pathways:

$$z'_m = \mathcal{O}_m(z_m; r, \mathcal{C}), \qquad m \in \{\text{data}, \text{policy}, \text{infer}\}. \tag{10}$$

For data synthesis, $(z_m, z'_m) = (\mathcal{D}, \mathcal{D}')$, and $\mathcal{O}_{\text{data}}$ curates, refines, or synthesizes training data. For policy learning, $(z_m, z'_m) = (\pi_\phi, \pi_{\phi'})$, and $\mathcal{O}_{\text{policy}}$ updates the policy. For inference-time verification, $(z_m, z'_m) = (\mathcal{Y}, y^*)$: $\mathcal{Y}$ is the set of candidate responses supplied to $\mathcal{O}_{\text{infer}}$, and y^* is the response selected or produced after ranking, verification, or revision.

This formulation treats the rubric as a model-agnostic intermediate representation within a complete construction–judgment–aggregation–optimization pipeline. The cri-

terion set $\mathcal{C}$ specifies *what* to evaluate, $\mathcal{J}$ evaluates each criterion, $\mathcal{F}$ forms the holistic reward, and $\mathcal{O} = \{\mathcal{O}_{\text{data}}, \mathcal{O}_{\text{policy}}, \mathcal{O}_{\text{infer}}\}$ determines how that reward and any explicit rubric guidance affect data synthesis, policy learning, or inference-time verification. Separating these components helps localize rubric-quality errors in $\mathcal{C}$, execution errors in $\mathcal{J}$, aggregation errors in $\mathcal{F}$, and optimization errors in $\mathcal{O}$, including cases where an apparently accurate reward fails to improve the downstream system or induces unintended behavior.

The interface $\mathcal{J}$ abstracts over the model-based evaluators and programmatic verifiers introduced in Section 2. It may use different evaluators for different criteria; the shared notation specifies their role in the pipeline rather than requiring a single judge architecture. This flexibility allows semantic judgments and deterministic checks to coexist within the same rubric.

Rubrics can also provide prospective guidance rather than only retrospective evaluation. In reward-only optimization, the policy observes only the reward derived from the criteria. Other methods additionally expose criteria or stage-specific feedback during generation, making intermediate objectives available to the policy (Zhou et al. 2025; Li et al. 2026a). These settings should be distinguished because they provide different information to the policy and may produce different behavior when rubric guidance is absent at inference time. Rubric RMs therefore play two complementary roles: they act as *measurement instruments* that produce rewards and, in some methods, as *explicit guidance* within $\mathcal{O}$. Evaluation should cover rubric quality, execution fidelity, aggregation validity, and downstream effectiveness rather than final preference accuracy alone.

Our computational formulation describes how rubric specifications are transformed into judgments, rewards, and downstream updates. To organize the literature, we examine the design questions that arise along this pathway. This motivates four complementary dimensions: **Foundations** examines what information is used to construct rubrics, how the rubrics are created and updated, and what evaluation criteria they contain; **Modeling** asks which components should be learned and how they should be trained; **Reasoning** asks how criteria should be interpreted, applied, and combined into an overall reward; and **Optimization** asks how the resulting signals should guide downstream learning and decision-making. These dimensions connect the computational framework to the methodological choices studied in existing work. In particular, Modeling cuts across multiple components, including the rubric generator and the judge, while Reasoning examines their evaluation behavior and aggregation mechanisms.

4. Foundations

The Foundations section focuses on the comprehensive pipeline of rubric construction, encompassing Rubric Inputs, Rubric Synthesis, and Rubric Content. Specifically, it addresses what information is fed into the generator, how and when the rubric is constructed, and the specific rubric content it comprises.

4.1 Rubric Inputs

The rubric construction process can be categorized into three settings based on how many candidate responses are considered jointly to extract the criteria: ***Point-wise***: The generator synthesizes a rubric by examining the query (Lv et al. 2026) or a single optional response (Bi et al. 2025) at a time to formulate absolute criteria, optionally accompanied by reference examples (Wang et al. 2025b). This manner is simple, but often suffers from calibration issues due to the lack of explicit comparison. ***Pair-wise***:

Table 1
Comparison of Representative Rubric Construction Approaches

Approach	Inputs	Synthesis	Content	Granularity
Static				
HealthBench (Arora et al. 2025)	-	Human designed	-	Instance & Dataset
ProfBench (Wang et al. 2025c)	-	Human designed	-	Instance
PaperBench (Starace et al. 2025)	-	Human designed	-	Instance
SALMON (Sun et al. 2024b)	-	Human designed	-	Dataset
Constitutional AI (Bai et al. 2022)	-	Human designed	-	Dataset
G-EVAL (Liu et al. 2023)	-	Human designed	-	Dataset
Kardia-R1 (Yuan et al. 2025a)	-	Human designed	-	Dataset
RaR (Gunjal et al. 2025)	Point-wise / Expert guidelines	Synthetic / Direct construct	-	Instance
RLCF (Viswanathan et al. 2025)	Point-wise	Synthetic / Direct construct	-	Instance
OpenRubrics (Liu et al. 2025a)	Pair-wise	Synthetic / Direct construct	-	Instance
CaRR (Zhang et al. 2026a)	Point-wise	Synthetic / Direct construct	Factual constraints	Instance
LEGIT (Lee et al. 2025)	Point-wise	Synthetic / Direct construct	Factual constraints	Instance
RGR-GRPO (Bi et al. 2025)	Point-wise	Synthetic / Direct construct	Outcome & Process	Instance
ORBIT (Wang et al. 2025b)	Point-wise / Factual info	Synthetic / Direct construct	Positive & Negative	Instance
RubricRAG (Dhole and Agichtein 2026)	Point-wise / Factual info	Synthetic / Direct construct	Positive & Negative	Instance
Ma et al. (2025)	Point-wise / Factual info	Synthetic / Direct construct	-	Instance
Chasing the Tail (Zhang et al. 2025a)	Pair-wise	Synthetic / Iterative refinement	-	Instance
RRD (Shen et al. 2026)	List-wise	Synthetic / Iterative refinement	-	Instance
ARR (Tian et al. 2026)	Pair-wise	Synthetic / Iterative refinement	-	Instance
Agentic Rubrics (Raghavendra et al. 2026)	Point-wise / Factual info	Synthetic / Iterative refinement	-	Instance
RLBFF (Wang et al. 2026c)	Point-wise / Expert guidelines	Synthetic / Filtering mechanisms	-	Instance
AutoRule (Wang and Xiong 2025)	Pair-wise	Synthetic / Filtering mechanisms	-	Dataset
RubricHub (Li et al. 2026c)	Point-wise / Expert guidelines	Synthetic / Filtering mechanisms	Verifiable & Semantic	Instance
Goel et al. (2025)	Point-wise	Synthetic / Filtering mechanisms	-	Instance
Liu et al. (2025c)	Listwise	Synthetic / Filtering mechanisms	-	Instance
Auto-Rubric (Xie et al. 2025)	Pair-wise	Synthetic / Iterative refinement & Filtering mechanisms	-	Dataset
Dynamic				
DR TuLu (Shao et al. 2025)	List-wise / Factual info	Synthetic / Filtering mechanisms	Positive & Negative	Instance
OnlineRubrics (Rezaei et al. 2025)	Pair-wise	Synthetic / Filtering mechanisms	-	Instance
RubricEM (Li et al. 2026a)	List-wise	Synthetic / Filtering mechanisms	-	Instance
ROPD (Fang et al. 2026)	Pair-wise	Synthetic / Direct construct	-	Instance
CaT (Jayalath et al. 2025)	Point-wise	Synthetic / Direct construct	-	Instance
RLCER (Sheng et al. 2026)	Point-wise	Synthetic / Filtering mechanisms	-	Instance
RLAC (Wu et al. 2025a)	Point-wise	Synthetic / Direct construct	-	Instance

The generator considers two responses jointly, extracting criteria based on the contrasting features that determine relative preference (Rezaei et al. 2025; Liu et al. 2026a; Xie et al. 2025). This manner naturally produces discriminative criteria and aligns with preference learning frameworks. However, because it is based on local comparisons,

it may not fully capture the relative ordering of a larger candidate set. ***List-wise***: The generator analyzes a broader set of responses simultaneously to distill criteria that govern their relative ordering and global quality (Shao et al. 2025; Shen et al. 2026; Li et al. 2026a; Huang et al. 2026). This manner can capture richer ranking information than pairwise comparison, though it is typically more complex and costly to implement.

The input to generator may include not only the response itself, but also context for rubric construction. Such context is particularly important in long-horizon decision-making tasks, where responses are often affected by hallucinations and cross-step inconsistencies. To improve rubric quality, prior work enriches the context with signals of different origins, ranging from model-internal knowledge to externally grounded feedback. These contextual signals can be broadly organized into the following categories:

Expert guidelines: A common way to enrich rubric context is to introduce normative signals designed by humans, such as predefined extraction principles (Gunjal et al. 2025; Li et al. 2026c; Yuan et al. 2025b) or annotations (Wang et al. 2026c; Lu et al. 2025). For example, RaR (Gunjal et al. 2025) manually defined the following extraction principles: Grounded in Expert Guidance, Comprehensive Coverage, Criterion Importance, and Self-Contained Evaluation, to help the model better derive rigorous and highly actionable rubrics. In contrast, RLBFF (Wang et al. 2026c) relies on human-written paragraph-length textual feedback to effectively improve the quality of the extracted rubrics and reduce hallucinations. These guidelines provide explicit standards and can substantially improve quality and consistency of the generated rubrics, especially in tasks requiring structured judgment. However, such context often comes with high construction cost and limited scalability, since the quality of the rubric depends on the availability of domain expertise and human supervision.

Factual information: Another line of work augments rubric context with grounded factual signals, including retrieved knowledge from external environments (Shao et al. 2025; Ma et al. 2025), retrieving domain knowledge and cases from semantically similar queries as reference (Wang et al. 2025b; Dhole and Agichtein 2026) or feedback from interactions with external systems such as agents or codebases (Raghavendra et al. 2026). DR TuLu (Shao et al. 2025) employs tools (such as Google Search and paper search) to retrieve real-world search snippets, utilizing this searched factual knowledge to generate rubrics. ORBIT (Wang et al. 2025b) proposes a Retrieval-Augmented Generation (RAG) framework that retrieves pre-built clinical cases and expert rubric databases to serve as few-shot examples, generating rubrics tailored for new diagnostic dialogues. Agentic Rubrics (Raghavendra et al. 2026) utilizes a Software Engineering agent equipped with tool-calling capabilities to explore the codebase, collecting contextual information such as specific functions and files to ultimately formulate codebase-specific rubrics. Compared with purely intrinsic context, this paradigm can more effectively mitigate hallucinations in open-domain and complex tasks by anchoring rubric construction to verifiable evidence. Its main limitation is that rubric quality becomes highly dependent on the reliability, latency, and coverage of the external feedback source.

4.2 Rubric Synthesis

Given a target task and input, the corresponding rubric can be constructed and incorporated into the alignment pipeline, which can be characterized along two dimensions: source and adaptability.

4.2.1 Source of Synthesis. Source of Synthesis concerns whether the rubric is constructed manually by humans or synthesized automatically by LLMs.

Human expert based methods rely on manually designed criteria provided by human experts. Such rubrics often encode subjective yet valuable human judgments, domain-specific standards, and task-specific requirements that are difficult to capture automatically, especially in specialized or high-stakes domains. Early work often focused on universal rubrics designed for a dataset or domain (Bai et al. 2022; Sun et al. 2024b), while more recent efforts increasingly emphasize instance-specific and fine-grained criteria (Arora et al. 2025; Starace et al. 2025; Wang et al. 2025c). Although expert-designed rubrics usually offer strong alignment quality and interpretability, they are inherently limited in scalability, since constructing and maintaining high-quality rubrics requires substantial human effort and expertise.

Synthetic methods construct rubrics automatically, typically using LLMs as the generator. A prevalent mechanism involves directly prompting LLMs to generate rubrics tailored to specific inputs (Gunjal et al. 2025; Viswanathan et al. 2025). Beyond simple one-shot prompting, recent methods also explore **iterative refinement** (Zhang et al. 2025a; Shen et al. 2026; Tian et al. 2026) and **filtering mechanisms** (Wang et al. 2026c; Goel et al. 2025; Liu et al. 2025c; Wang and Xiong 2025) to extract more discriminative, reliable, and task-aligned criteria.

For **iterative refinement**, Zhang et al. (2025a) employ a progressive, multi-round workflow to capture the nuances of high-quality responses. In each iteration, they first score all candidate responses using the current rubric to identify the top two responses. A proposer LLM is then prompted to compare this single top-tier pair, extract distinguishing features, and directly encode them to refine the existing rubric. This updated rubric is then used to re-score the candidate pool in the subsequent round, progressively focusing on higher-quality responses and continuously refining its discriminative power round by round. Similarly, RRD (Shen et al. 2026) recursively decomposes coarse rubrics into finer-grained, discriminative criteria, which expands evaluation coverage while sharpening the separation between responses. For **filtering mechanisms**, AutoRule (Wang and Xiong 2025) directly prompts an LLM to merge and compress the extracted rule sets, eliminating duplicates to synthesize a compact, query-agnostic dataset-level core rubric set. RLBFF (Wang et al. 2026c) filters extracted principles by removing rubrics unsupported by human-written textual feedback and partially fulfilled rubrics and retain only high-precision, non-redundant criteria. Furthermore, Auto-Rubric (Xie et al. 2025) integrates both iterative refinement and filtering mechanisms. It first extracts a large-scale pool of query-specific rubrics through an iterative Propose-Evaluate-Revise pipeline and subsequently compresses this granular pool into a dataset-level core rubric set by maximizing an information-theoretic coding rate to eliminate redundancy. Taking a different approach, Huang et al. (2025) introduce a unique Rubric-First strategy, which pre-constructs a high-quality rubric library and subsequently filters or synthesizes queries that align with these rubrics.

Benefiting from their broad prior knowledge and generalization ability, LLMs can synthesize reasonably accurate rubrics across a wide range of tasks and domains, substantially improving scalability over purely human-designed approaches. However, the quality of synthetic rubrics still depends on the model's internal knowledge, prompt design, and external grounding signals when available. In particular, Orcust (Lu et al. 2025) takes a hybrid approach that integrates human-expert design with automatic synthesis. While human experts define explicit domain requirements and prohibited behaviors, LLMs are leveraged to synthesize articulate additional strategies. Despite the proliferation of work on rubric synthesis methods, conducting a fair comparison of these approaches remains a pressing challenge. Fortunately, recent work (Zhou and Tan 2026) has begun to consolidate these fragmented generation, refinement, and evaluation

into a composable pipeline, which enables a more systematic and rigorous comparison of different methodologies.

4.2.2 Adaptability of Synthesis. Beyond the source of synthesis, another important dimension within the Rubric RM framework is whether the rubrics, which serve as the foundational criteria for generating reward signals to guide the response generation policy, remain fixed during the reinforcement learning optimization process. Accordingly, existing approaches can be divided into static and dynamic paradigms.

Static synthesis generates a fixed set of rubrics prior to the reinforcement learning process, making them entirely independent of the evolving policy distribution or rollout behavior. Furthermore, the rubric generator itself remains frozen and does not adapt as reinforcement learning proceeds. This paradigm is simple, efficient, and easy to scale, and it remains the default setting in most works (Li et al. 2026c; Wang et al. 2026c). However, as the policy model π_ϕ evolves, fixed rubrics may gradually become incomplete or exploitable, failing to cover newly emerged failure modes. As a result, static criteria are especially vulnerable to Goodhart-like reward hacking, where the policy learns to satisfy superficial rubric signals without achieving the intended behavior. Some works attempt to alleviate this issue by manually summarizing observed failure patterns and injecting defensive criteria (Viswanathan et al. 2025; He et al. 2025), but such patching remains labor-intensive and difficult to sustain under continuous policy shift.

Dynamic synthesis updates, extracts, or regenerates rubrics online according to the current policy distribution or rollout behavior. Compared with static rubrics, this paradigm is better suited to capturing evolving failure modes and maintaining evaluative relevance throughout optimization. Recent methods explore several directions, including online rubric extraction, which dynamically constructs rubrics based on rollout results during the reinforcement learning process (Shao et al. 2025; Rezaei et al. 2025; Li et al. 2026a; Jayalath et al. 2025; Fang et al. 2026), and joint optimization frameworks where the policy model co-evolves with a dynamic rubric generator (Sheng et al. 2026; Wu et al. 2025a).

OnlineRubrics (Rezaei et al. 2025) elicits new evaluation criteria dynamically through pairwise comparisons between responses from the current policy and a control model (e.g., a reference policy). This pairwise online elicitation allows the reward model to actively penalize nuanced reward-hacking behaviors and capture emergent desirable qualities that static, point-wise rubrics often overlook. Furthermore, DR Tulu (Shao et al. 2025) introduces Reinforcement Learning with Evolving Rubrics specifically tailored for open-ended, long-form deep research tasks. At each training step, the framework contrasts multiple agent rollouts and uses retrieved external knowledge to generate new rubrics. To maintain efficiency and discriminative power, it manages an active rubric buffer by filtering out criteria with zero reward variance, retaining only those that effectively distinguish trajectory quality as the policy's distribution shifts.

RLCER (Sheng et al. 2026) focuses on Reinforcement Learning with CoT Supervision via Self-Evolving Rubrics. The policy model serves a dual purpose within this framework: it acts both as the problem solver generating the reasoning steps, and as the rubric generator establishing the criteria. It autonomously proposes natural-language scoring criteria to reward and refine its own chain of thought, while also self-evolving over time to continuously enhance its ability to generate high-quality rubric. RLAC (Wu et al. 2025a) frames free-form LLM optimization as an adversarial game. A learned rubric generator dynamically proposes a specific rubric (a likely failure mode) for the policy's output, which is then verified by an external validator. If the output fails,

the generator is rewarded; if it passes, the policy is rewarded. This avoids exhaustive enumeration and dynamically targets high-risk errors.

Although dynamic approaches are more promising for robust long-term optimization, they also introduce additional complexity in stability, computational cost, and coordination between rubric generation and policy learning.

4.3 Rubric Contents

Synthesized rubrics specify the criteria used to evaluate a response, action sequence, or decision trajectory. For complex tasks, rubrics are often designed with distinct evaluation foci to assess different aspects of performance separately. RGR-GRPO (Bi et al. 2025) focuses on the dual verification of **Outcome and Process**. The framework divides the rubric criteria into factual and process criteria. Factual criteria verify the accuracy of intermediate steps, sub-answers, and final results, while process criteria evaluate the validity of the reasoning path. This design ensures that the model not only arrives at the correct answer but also guarantees that the approach taken to reach it is reasonable. RubricHub (Li et al. 2026c) focuses on the objective and subjective distinction between **Verifiable and Semantic** criteria. Verifiable criteria cover objective constraints (e.g., format, word count) and are evaluated by rule-based systems. Conversely, semantic criteria address complex qualitative attributes like reasoning depth and tone, requiring LLM-based evaluation for deeper understanding. DR Tulu (Shao et al. 2025) and RubricRAG (Dhole and Agichtein 2026) jointly focus on **Positive and Negative** criteria. Positive rubrics encourage desired exploration and strength leveraging, while negative rubrics penalize undesirable behaviors and reward hacking.

Furthermore, rubrics can also serve as fine-grained **Factual Constraints** on both the content and the reasoning trajectory of response. LEGIT (Lee et al. 2025) illustrates this by structuring rubrics as hierarchical decision trees derived from expert judgments. Rather than solely verifying the final outcome, these rubrics act as strict step-by-step checkpoints, measuring whether the response has identified the legal issues mentioned in the court judgment. Similarly, CaRR (Zhang et al. 2026a) demonstrates this in the context of multi-hop search agents by utilizing atomic factual statements as evaluation rubrics. These rubrics impose fine-grained constraints throughout the agent's exploration phase by dictating that intermediate entities must be explicitly grounded in retrieved evidence and ensuring that these verified facts form an unbroken, logical chain leading to the final predicted answer.

At the granularity level, some studies (Xie et al. 2025; Wang and Xiong 2025) focus on the generalization of rubric content, exploring whether rubrics can transcend the instance-level to become common values at the dataset-level. Evaluating all data within a specific domain using a uniform rubric yields significantly stronger generalizability compared to instance-level rubrics, which are strictly tailored to individual instances. We further discuss this in Section 10. There is also work (Arora et al. 2025) that attempts to combine fine-grained, instance-level criteria with a set of dataset-level "consensus criteria," offering the advantage of balancing broad evaluation coverage for diverse contexts with rigorous constraints on universally critical behaviors. Despite these explorations, there is still no unified consensus or theoretical research on what rubric design is universally optimal; in practice, the most effective formulation often depends on the task complexity, domain requirements, and downstream optimization objective.

5. Modeling

The modeling stage focuses on learning the key components of the rubric evaluation pipeline, including rubric construction, rubric-guided judgment, and, in some cases, their joint optimization.

5.1 What to learn?

A rubric serves as an intermediate variable in rubric reward modeling, bridging the designer's intended evaluation objectives and downstream supervision signals. Accordingly, modeling in *Rubric RMs* involves two closely related goals: *optimizing the generator to construct high-quality rubrics*, and *optimizing the judge to perform reliable rubric-guided judgment*. Some studies focus on the *generator*, which serves as the core component of *Rubric RMs* by constructing task-specific evaluation criteria (Lv et al. 2026; Fan et al. 2026; Dhole and Agichtein 2026). Lv et al. (2026) train a generator aligned with human values using human preference data.

In contrast, other studies target the criterion-level *judge* $\mathcal{J}$ to improve its ability to interpret generated rubrics and apply each criterion reliably (Ma et al. 2025; Hashemi et al. 2024). Aggregation $\mathcal{F}$ remains a conceptually separate component, although implicit-aggregation methods may absorb it into the judge. LLM-RUBRIC (Hashemi et al. 2024) aligns the judge's evaluations with human values by training a personalized calibration network. Beyond such separate optimization, Xu et al. (2026a) simultaneously optimize the generator and the judge, while Chen et al. (2025b) and Liu et al. (2025c) treat rubric construction and rubric-guided judgment as a unified process, enabling end-to-end optimization.

5.2 How to Learn?

In the absence of explicit intermediate supervision for rubric construction and judgment, many studies directly leverage a strong LLM as the generator or the judge, relying on its pretrained capabilities for rubric synthesis and score aggregation (Li et al. 2026c). While this training-free strategy is flexible, it often struggles in complex scenarios that demand hierarchical intent structures, faithful subtask decomposition, and task-specific reasoning. To mitigate these weaknesses, some works adopt SFT (Liu et al. 2025a; He et al. 2025; Liu et al. 2026a), which uses large amounts of high-quality training data to teach the model to follow the desired rubric format and perform more reliable task planning and decomposition. More advanced approaches introduce reinforcement learning based methods, which alleviate the reliance on explicit supervision over intermediate reasoning processes and instead optimize model behavior directly from preference signals. Typical examples include Direct Preference Optimization (DPO) (Wu et al. 2025a) and Group Relative Policy Optimization (GRPO) (Lv et al. 2026; Dhole and Agichtein 2026; Chen et al. 2025b; Fan et al. 2026). For example, Lv et al. (2026) leverage GRPO with a hybrid reward function designed to align rubric generation directly with human preferences, which consists of three complementary components. The preference consistency reward acts as the core signal, assigning a positive reward if the generated rubric correctly yields a higher score for a human-preferred report compared to a rejected one. This is further stabilized by an LLM-as-a-judge reward that evaluates the intrinsic quality of the rubric (e.g. logical coherence, comprehensive coverage) and a hard format reward that penalizes non-compliance with the required JSON schema.

Distinctively, MR-RML (Jin et al. 2025a) internalizes multidimensional medical criteria directly into an independent judge by utilizing geometric projection reference constraints. CROME (Srivastava et al. 2026) constructs preference data by isolating individual criteria and trains the judge to internalize this knowledge directly into its parameters. Both approaches inject rubric knowledge into the judge's parameters, enabling the evaluation of inputs without relying on explicit rubrics generated in real-time, which significantly reduces computational overhead. However, this internalization effectively renders the judge a black box.

Table 2
Comparison of Representative Rubric RM Reasoning Approaches

Approach	Scoring Scale	Aggregation Paradigm	Interp.
AutoRule (Wang and Xiong 2025)	{0,1}	Explicit / Unweighted Linear	✓
RubricRL (Feng et al. 2025)	{0,1}	Explicit / Unweighted Linear	✓
DR Tulu (Shao et al. 2025)	{0,0.5,1}	Explicit / Weighted Linear	✓
RRD (Shen et al. 2026)	{0,1}	Explicit / Weighted Linear	✓
OnlineRubrics (Rezaei et al. 2025)	{0,1}	Explicit / Weighted Linear	✓
RubricHub (Li et al. 2026c)	{0,1}	Explicit / Weighted Linear	✓
RLCF (Viswanathan et al. 2025)	0-100	Explicit / Synergy with Rule RMs	✓
VERIF (Peng et al. 2025)	{0,1}	Explicit / Synergy with Rule RMs	✓
Orcust (Lu et al. 2025)	1-10	Explicit / Synergy with Rule RMs	✓
Huang et al. (2025)	1-5	Explicit / Synergy with Rule RMs & Structure-aware	✓
AdvancedIF (He et al. 2025)	{0,1}	Explicit / Structure-aware	✓
RGR-GRPO (Bi et al. 2025)	{0,1}	Explicit / Structure-aware	✓
RaR (Gunjal et al. 2025)	1-10	Implicit	×
OpenRubrics (Liu et al. 2025a)	Preference signal	Implicit	×
Kimi K2 (Bai et al. 2025a)	Preference signal	Implicit	×
Yuan et al. (2025b)	0-10	Implicit	×
SALMON (Sun et al. 2024b)	Preference signal	Implicit	×

6. Reasoning

Rubric Reasoning focuses on how multiple criteria are jointly considered by the judge to produce the final reward or supervision signal. This process is inherently challenging because **the relationships among criteria can be complex**: they may be *complementary*, *partially overlapping*, *hierarchically dependent*, or even *conflicting*. Therefore, rubric reasoning is not merely a matter of simple aggregation, but of deciding how these criteria should interact, be prioritized, and be balanced in context.

Aggregation defines reward semantics, not only numerical format. A weighted sum assumes independent and compensatory criteria, whereas alignment objectives may contain prerequisites or veto relations. Thus, identical criterion-level judgments can induce different policy behavior under different aggregation rules, making these assumptions central to holistic reward formation and credit assignment.

6.1 Explicit Aggregation

Explicit aggregation maps criterion-based judgments over a selected rubric set $\mathcal{C}$ into a final scalar value through an explicit aggregation function. Concretely, the judge first assigns a score s_i, usually a binary score, to each criterion $c_i \in \mathcal{C}$, yielding a score vector $\mathbf{s} = [s_1, \ldots, s_K]^\top$, which is then converted into a scalar reward r by applying $\mathcal{F}(\mathbf{s})$. It is worth noting that the evaluation signal representation, denoted as s_i, lacks a unified standard. Existing works range from assigning point-wise scalar scores such as absolute binary judgments (e.g., 0 or 1) and broader numerical scales (e.g., 1-5, 1-10, or 0, 0.5, 1) to adopting pairwise preference signals (e.g., binary preference labels indicating "A is better than B") to capture richer criterion-level feedback. Depending on the relational assumptions encoded in $\mathcal{F}$, such aggregation functions can be broadly divided into two primary categories: *linear aggregation*, which assumes additive and largely independent contributions across criteria, and *structure-aware aggregation*, which captures richer interactions such as hierarchical dependencies.

Linear Aggregation. A linear function provides a simple and interpretable way to combine multiple evaluation dimensions: $\mathcal{F}(\mathbf{s}) = \frac{\sum_i w_i s_i}{\sum_i w_i}$. Here, w_i reflects the relative importance of criterion c_i in the final evaluation. A larger weight indicates a stronger contribution of the corresponding dimension, whereas a smaller weight suggests lower importance. In many cases, such weights must be manually specified or calibrated with additional supervision in order to faithfully reflect the relative importance of different criteria (Gunjal et al. 2025; Shen et al. 2026). Since such supervision is often unavailable in practice, many approaches adopt uniform weighting (Wang and Xiong 2025; Feng et al. 2025), e.g., $w_i = 1$, which implicitly assumes that all criteria are equally important. While this simplification improves usability, it also imposes a strong assumption and may limit the generality of linear aggregation in complex scenarios. Furthermore, some works (Peng et al. 2025; Viswanathan et al. 2025; Lu et al. 2025) target scenarios where hard constraints (e.g., format) and soft constraints (e.g., logic) coexist, thereby achieving a synergy between Rule RMs and Rubric RMs. Overall, linear aggregation is interpretable and flexible, making it suitable for dynamically updating rubrics to mitigate reward hacking. However, it struggles to capture complex non-linear dependencies and its computational cost typically grows linearly with the number of criteria.

Structure-aware Aggregation. In complex settings, criteria may exhibit non-linear trade-offs or hierarchical dependencies (e.g., safety typically possesses veto power), meaning that the contribution of one dimension to the final reward depends on the values of the others. To address this, Huang et al. (2025) attempt to employ hand-crafted non-linear rules to capture the complex relationships among various criteria. Similarly, He et al. (2025) assign a reward signal if and only if all criteria are strictly satisfied. Furthermore, Bi et al. (2025) implement a dynamic strategy that activates the evaluation of subsequent process criteria only upon the successful verification of factual outcomes. Despite these advancements, relying on such manually hard-coded logic still struggles to fully encompass complex interactions.

6.2 Implicit Aggregation

Implicit Aggregation leverages the powerful in-context learning (ICL) capabilities of LLMs. This method foregoes the intermediate step of item-by-item scoring. Instead,

it inputs the entire rubric $\mathcal{C}$ as a context prompt into the model. The model performs internal reasoning to weigh various criteria, directly outputting a final scalar signal. The judge $\mathcal{J}$ implicitly subsumes the roles of both individual criterion scoring and the explicit aggregation function $\mathcal{F}$. Gunjal et al. (2025) first coined the term Implicit Aggregation and demonstrated experimentally that it outperforms Explicit Aggregation under equivalent conditions. Following this trajectory, Yuan et al. (2025b), Bai et al. (2025a), and Liu et al. (2025a) have also adopted this approach. Despite its empirical success in capturing complex interactions, Implicit Aggregation presents significant challenges. Its inherent "black-box" nature severely restricts interpretability and fine-grained controllability, as researchers can neither observe intermediate criteria scores nor directly adjust their weights. Furthermore, this approach is hindered by context length constraints, increased prefill costs for extensive rubrics, and susceptibility to position bias, where the ordering of criteria within the prompt can skew the model's judgment.

7. Rubric RMs as the Core of the Optimization Loop

Rubric RMs play a central role in the optimization loop of downstream tasks by providing both *supervision* and *control* signals. Beyond assigning scalar rewards, they improve data quality, guide policy optimization, and strengthen inference-time decision making by translating implicit and holistic task objectives into explicit and analyzable criteria. In the following, we discuss how *Rubric RMs* contribute to downstream optimization across different stages of the loop.

7.1 Data Synthesis

Compared with traditional data synthesis methods, rubrics enable more fine-grained quality control over both synthesized outputs and their intermediate construction processes via structured criteria. For result quality, RubricHub (Li et al. 2026c) develop automated coarse-to-fine rubric generation frameworks to construct comprehensive, fine-grained, and highly discriminative rubrics, and further use rubric-based rejection sampling to curate fine-tuning data. To further optimize data synthesis, Optimsyn (Fan et al. 2026) treat rubric generation as a learnable policy. It utilizes influence-guided optimization to generate rubrics and synthesize corresponding data, thereby maximizing the synthesized data's training utility on the target model. For process control, rubrics can additionally serve as iterative guidance signals in long-horizon generation. For example, ARISE (Wang et al. 2025e) shows that rubrics not only evaluate intermediate outputs, but also drive step-by-step refinement throughout agentic optimization.

7.2 Policy Learning

Policy learning is the most widely represented application of Rubric RMs in the literature surveyed here, as summarized in Table 3: rubrics are converted into reward signals that supervise policy updates in open-ended tasks where a single verifiable outcome is often unavailable. Rubric-based policy learning can be understood by comparing *how rubric-derived signals are designed and used within the optimization loop*. Existing methods can be characterized along four complementary and non-exclusive axes: what part of the trajectory is evaluated and at what granularity credit is assigned; where the policy is coupled to the reward pipeline, including rubric construction and judgment; how the reward pipeline improves supervision quality through adaptive rubric synthesis or

Table 3
Comparison of Rubric RM Applications in the Optimization Loop

Approach	Description	Domain
Data Synthesis		
RubricHub (Li et al. 2026c)	Rubric-based Rejection Sampling Fine-Tuning to synthesize high-quality SFT data.	Multi-domain
Optimsyn (Fan et al. 2026)	Dynamically optimizes the rubric generator via RL to guide the synthesis of SFT data.	Multi-domain
Kardia-R1 (Yuan et al. 2025a)	Employs rubric to provide feedback for iterative refinement, synthesizing multi-turn empathetic dialogue data for SFT and RL.	Emotional dialogues
ARISE (Wang et al. 2025e)	Employs a rubric-guided multi-agent iterative refinement loop to assess and systematically enhance survey drafts.	Science
rDPO (Yu et al. 2026b)	Employs instance-specific rubrics to filter on-policy responses and synthesize high-quality preference pairs for iterative DPO.	Multimodal
Policy Learning		
Lv et al. (2026)	Applies outcome-level scalar rewards computed by LLM-as-a-judge on generated rubrics to supervise policy optimization via GRPO.	DeepResearch
CaT (Jayalath et al. 2025)	Applies outcome-level rewards by scoring final rollouts against self-proposed rubrics generated from a synthesized reference.	Multi-domain
RubricRL (Feng et al. 2025)	Applies outcome-level rewards by evaluating generated images against prompt-adaptive rubrics via GRPO.	Multimodal
CaRR (Zhang et al. 2026a)	Evaluates complete search trajectories with citation-aware rubrics and aggregates their satisfaction into a trajectory-level reward.	DeepResearch
LEGIT (Lee et al. 2025)	Scores legal issues for coverage and correctness, then aggregates them with final-order correctness into a trajectory-level reward.	Law
RTT (Xu et al. 2026b)	Applies fine-grained token-level rewards by mapping coarse rubric scores to specific tokens and optimizes the policy via RTT-GRPO.	General
Ye et al. (2025)	Utilizes the policy model itself as a rubric-guided judge to evaluate open-ended responses via GRPO.	Medical
EvoLM (Li et al. 2026b)	Alternately optimizes parameter-shared policy and rubric-generator roles, while a separate frozen judge provides rubric-conditioned rewards.	Multi-domain
RLCER (Sheng et al. 2026)	Jointly optimizes parameter-shared reasoner and rubricator roles, while a frozen verifier aggregates rubric satisfaction into a CoT-level reward.	Multi-domain
DR Tulu (Shao et al. 2025)	Dynamically generates rubrics from current policy rollouts and maintains a variance-filtered buffer to keep supervision discriminative.	DeepResearch
Liu et al. (2025c)	Scales reward-model judgment by sampling critiques and predictions, filtering with a meta RM, and aggregating retained predictions by voting.	General
RuscaRL (Zhou et al. 2025)	Employs rubrics as both explicit scaffolding for exploration and verifiable rewards to alleviate the exploration bottleneck.	General
RubricEM (Li et al. 2026a)	Uses self-generated rubrics to guide planning, iterative evidence gathering, review, and synthesis, while SS-GRPO assigns stagewise credit.	DeepResearch
RGR-GRPO (Bi et al. 2025)	Integrates rubric-based fine-grained rewards and uses rubrics to guide off-policy refinement for broader exploration.	Multi-domain
HeRL (Zhang et al. 2026c)	Leverages failed trajectories and unmet rubrics as hindsight experience to guide exploration, adding a bonus reward for improvement potential.	Multi-domain
THINK-WITH-RUBRICS (Yu et al. 2026a)	Internalizes rubric generation into the reasoning trajectory, jointly optimizing with golden and self-generated rubric consistency rewards.	General
Inference-time Verification		
DeepVerifier (Wan et al. 2026)	Employs rubric-guided verification to decompose complex verification into sub-questions, providing iterative feedback for self-correction.	DeepResearch
Agentic Rubrics (Raghavendra et al. 2026)	Generates context-grounded rubrics via agentic codebase exploration to serve as scalable, execution-free verifiers for code patches.	Coding

scaled judgment; and whether rubrics influence the policy only through reward signals or are also exposed as explicit guidance during generation.

The first and most fundamental axis concerns *credit assignment*, namely where rubric-derived feedback is attached within a generated trajectory. Outcome-level reward remains the most widely adopted design: it evaluates a completed response or rollout against multiple criteria and typically reduces these judgments to a rollout-level reward (Lv et al. 2026; Shao et al. 2025; Jayalath et al. 2025). This design is simple to

integrate with policy optimization, but it leaves temporal credit assignment unresolved. Criterion-level decomposition identifies which quality dimensions are satisfied, not which states, reasoning steps, or tokens caused the resulting score; even a vector of criterion scores therefore remains sparse response-level supervision unless its components are attributed to particular decisions. Importantly, *process-level* is used in two senses: evaluating the reasoning process and assigning feedback within that process, and only the latter directly improves temporal localization. Methods that inspect reasoning traces or tool-use trajectories can provide more process-sensitive supervision (Yuan et al. 2025b; Zhang et al. 2026a; Lee et al. 2025; Sheng et al. 2026), yet they may still aggregate those judgments into one trajectory-level reward. More direct temporal localization is explored by stage-structured methods that assign feedback to phases of long-horizon trajectories (Li et al. 2026a) and token-level methods that map response-level criteria to responsible tokens (Xu et al. 2026b). Such signals can be denser and more localized, but their usefulness depends on reliable attribution; otherwise, intermediate errors can propagate into policy updates, while additional stage- or token-level evaluation increases computational cost. Choosing the supervision granularity therefore remains a trade-off among localization, attribution reliability, efficiency, and training stability.

Second, methods differ in the locus of coupling between the policy and the reward pipeline. Most approaches keep the policy separate from rubric construction and rubric-guided judgment. Self-rewarding methods instead reuse the policy as a rubric-guided judge (Ye et al. 2025), coupling the policy directly to judgment. EvoLM (Li et al. 2026b) and RLCER (Sheng et al. 2026) couple the policy at a different interface: response generation and rubric construction are instantiated as roles of a parameter-shared model, while rubric satisfaction is assessed by a separate frozen judge or verifier. Thus, self-rewarding internalizes judgment, whereas EvoLM and RLCER internalize rubric construction without making the policy its own judge.

Third, methods differ in how the reward pipeline improves the quality of supervision delivered to policy optimization. One route enhances rubric construction. As discussed under dynamic synthesis in Section 4.2.2, DR Tulu dynamically generates and maintains rubrics from current policy rollouts, retaining criteria that remain discriminative as policy behavior evolves. Another route enhances rubric-guided judgment. Liu et al. (2025c) increase reward-model inference compute by sampling multiple principle-conditioned critiques and reward predictions, using a meta reward model to filter unreliable samples, and aggregating the retained predictions through voting. The former keeps the reward specification aligned with the current policy distribution, whereas the latter improves the reliability of each reward computation.

Finally, rubrics differ in their functional role within learning. Most methods expose the policy only to aggregated rubric-derived rewards. Other methods additionally make the criteria available during generation or use them to organize the learning trajectory. RuscaRL (Zhou et al. 2025) injects rubric scaffolds into rollout instructions to expand exploration and gradually removes this support, while RubricEM (Li et al. 2026a) conditions distinct research stages on rubrics and assigns stagewise feedback for long-horizon optimization. Comparing these designs therefore requires more than final task performance; relevant considerations include learning efficiency, exploration ability, training stability, and robustness to reward hacking and distribution shift.

7.3 Inference-time Verification

At inference time, *Rubric RMs* can serve as a verifier to directly improve policy outputs, extending its role beyond training supervision into inference-time decision making.

They can support ranking, verification, and revision of candidate outputs, for example in Best-of-N, reranking, self-refinement, and critique-and-revision pipelines, by providing explicit criteria for test-time evaluation. For instance, Wan et al. (2026) propose DeepVerifier, a rubric-guided verification framework for Deep Research Agents that enables iterative self-correction through test-time feedback and retry. Beyond iterative refinement, Agentic Rubrics (Raghavendra et al. 2026) show that context-grounded rubrics constructed by agents can serve as an effective and scalable execution-free verifier for SWE agents, improving test-time selection while providing more interpretable reward signals.

7.4 Domain Adaptation

Rubrics are naturally applicable across a wide range of domains because many downstream tasks rely on implicit expert knowledge that cannot be easily captured by preference signals. The core value of rubric modeling lies in making such domain knowledge explicit by recovering it from task requirements and data, providing a more scalable way to align models with domain-specific objectives. This is particularly important in domains such as medicine (Arora et al. 2025; Wang et al. 2025b; Jin et al. 2025a), law (Lee et al. 2025), science (Goel et al. 2025), mathematics (Yuan et al. 2025b), and coding (Raghavendra et al. 2026), where outputs must satisfy fine-grained constraints over multiple criteria. More broadly, in complex settings such as DeepResearch (Shao et al. 2025; Li et al. 2026a; Zhang et al. 2026a; Lv et al. 2026), and multimodal reasoning (Lu et al. 2025; Feng et al. 2025), rubrics can further incorporate expert priors, tool feedback, and environmental interactions as structured constraints, enabling models to better coordinate decision making under real-world task requirements.

8. Training Dataset for Rubric RMs

As shown in Table 4, we summarize the high-quality data resources currently used for training, including datasets with explicit rubrics and source datasets used to extract criteria.

Rubric Datasets. Recent works have focused on extracting or constructing reusable rubric datasets to facilitate fine-grained evaluation and alignment. RUBRICHUB is a large-scale, multi-domain rubric dataset designed to evaluate and optimize open-ended generation by providing fine-grained, highly discriminative supervision. WILDCHECKLISTS provides a large-scale collection of synthetically generated checklists to evaluate fine-grained constraint satisfaction using LLM judges and verifier programs, facilitating language model alignment through reinforcement learning. OPENRUBRICS is a large-scale, diverse collection of synthetic prompt and rubric pairs derived through contrastive generation. In specialized domains, Gunjal et al. (2025) introduce RAR-MEDICINE and RAR-SCIENCE consisting of LLM-generated, instance-specific evaluation criteria, which are utilized as structured, multi-dimensional reward signals for on-policy reinforcement learning to improve complex reasoning in expert domains. LEGIT introduces expert-level rubrics structured as hierarchical legal issue trees extracted from real-world court judgments, which are used to systematically evaluate LLM-generated reasoning traces for both issue coverage and logical correctness. Ultimately, these consolidated rubric datasets serve multiple critical functions: training automated rubric generators, providing dense reward signals for reinforcement learning, and filtering high-quality training data.

Table 4
Overview of Dataset.

Dataset	Domain	Modality	Scale
Rubric Dataset			
RUBRICHUB (Li et al. 2026c)	*Multi-domain*	Language	110k
WILDCHECKLISTS (Viswanathan et al. 2025)	*Multi-domain*	Language	130K
OPENRUBRICS (Liu et al. 2025a)	*Multi-domain*	Language	35.6k
RAR-MEDICINE (Gunjal et al. 2025)	*Medical*	Language	20k
RAR-SCIENCE (Gunjal et al. 2025)	*Science*	Language	20k
LEGIT (Lee et al. 2025)	*Legal*	Language	24K
Source Dataset			
WILDCHAT (Zhao et al. 2024)	*Multi-domain*	Language	1m
ULTRAFEEDBACK (Cui et al. 2024)	*Multi-domain*	Language	64k
SKYWORK-REWARD (Liu et al. 2024)	*Multi-domain*	Language	80k
HELPSTEER3-PREFERENCE (Wang et al. 2025d)	*Multi-domain*	Language	40k
MAGPIE (Xu et al. 2025)	*Multi-domain*	Language	300k
MEGASCIENCE (Fan, Wang, and Liu 2025)	*Multi-domain*	Language	1.25m
LONGWRITER (Bai et al. 2025b)	*Writing*	Language	6k
LONGWRITER-ZERO (Wu et al. 2025b)	*Writing*	Language	8.6k
LONGALIGN (Bai et al. 2024)	*Writing*	Language	10k
LMSYS (Zheng et al. 2024)	*Chat*	Language	1m
MEDICAL-O1 (Chen et al. 2024)	*Medical*	Language	19.7k
RLAIF-V (Yu et al. 2025)	*Multi-domain*	Multi-modal	83.1k
LLAVA-HUMAN-PREFERENCE-10K (Sun et al. 2024a)	*Multi-domain*	Multi-modal	9.4k
LLAVA-CRITIC-113K (Xiong et al. 2025)	*Multi-domain*	Multi-modal	113.0k
MM-RLHF (Zhang et al. 2025d)	*Multi-domain*	Multi-modal	16.3k
PIXMO-CAP (Deitke et al. 2025)	*Multi-domain*	Multi-modal	717k
DENSEFUSION-1M (Li et al. 2024b)	*Multi-domain*	Multi-modal	1m
VIRL-39K (Wang et al. 2025a)	*Multi-domain*	Multi-modal	39k

Source Datasets. Unlike rubric datasets, these resources do not necessarily contain explicit criteria; instead, they provide prompts, responses, or preferences from which rubrics and rubric-guided supervision can be constructed. Existing rubric construction processes typically rely on adapting various source and preference datasets. For instance, WILDCHAT provides a large-scale, multilingual corpus of real-world user-chatbot interactions to capture natural user behaviors. To facilitate preference modeling, datasets like ULTRAFEEDBACK and SKYWORK-REWARD provide large-scale, automated preference data; the former utilizes GPT-4 for fine-grained scores and critiques, while the latter offers lightweight, high-quality filtered pairwise comparisons from public sources. Notably, while these rely heavily on model-generated feedback or automated filtering, HELPSTEER3-PREFERENCE distinguishes itself by providing high-quality, human-annotated preference pairs. For a complete list of source datasets, please refer to Table 4.

9. Evaluation and Benchmarking

Current evaluation of *Rubric RMs* follows two substantially different regimes. A small body of work directly examines one component of the reward pipeline, whereas most

studies report final judgment accuracy or downstream policy performance and therefore evaluate rubric construction, judge behavior, aggregation, and policy learning jointly. Such end-to-end results establish system utility, but cannot identify which component produced either the gain or the failure.

Accordingly, we distinguish four interface-aligned diagnostic targets: **rubric quality**, **execution fidelity**, **aggregation validity**, and **downstream effectiveness**. They ask whether the rubric represents the intended objective, the judge applies its criteria faithfully, the aggregation implements the intended decision semantics, and the rubric-mediated signal produces the intended effects in data synthesis, policy learning, or inference-time verification. These targets are conceptually distinct but not inherently independent: ambiguous criteria can induce judge errors, and implicit aggregation can make $\mathcal{J}$ and $\mathcal{F}$ observationally inseparable. Component-level attribution is therefore justified only when the relevant intermediate variables are exposed and suitable controls or reference signals are available. Evaluating $\mathcal{C}$ requires holding $\mathcal{J}$ and $\mathcal{F}$ fixed wherever possible; evaluating $\mathcal{J}$ additionally requires criterion-level reference judgments under a fixed rubric; evaluating $\mathcal{F}$ requires fixed criterion-level judgments and an independent decision standard; and evaluating downstream effectiveness requires coupling the signal to the relevant downstream process and measuring external outcomes.

Table 5 follows a hierarchical classification rule. *Component-level meta-evaluation* benchmarks take precedence because they directly diagnose a specific component of the Rubric RM pipeline. Among the remaining resources, *rubric-based benchmarks* provide structured rubrics or scenario-specific criteria as a central, released scoring artifact or evaluation protocol, whereas *general reward and judge benchmarks* center their evaluation interface on holistic preferences, ratings, or response selection. Rubric-based benchmarks removes variation from $\mathcal{G}$ by using a fixed rubric and holding $\mathcal{C}$ constant, but does not by itself separate $\mathcal{J}$ from $\mathcal{F}$: when only final scores or preferences are observed, the benchmark evaluates their composite effect $\mathcal{F} \circ J$. Direct evaluation of $\mathcal{J}$ additionally requires criterion-level reference judgments, whereas direct evaluation of $\mathcal{F}$ requires fixed criterion-level scores and aggregation-specific targets such as an expert decision policy, a validated holistic ordering, or explicit priority and veto constraints. General reward and judge benchmarks can instead provide indirect evidence about generated rubrics by applying the complete rubric–judge–aggregation pipeline and comparing its outputs with benchmark labels. Such end-to-end results remain useful for system-level validation, but cannot attribute performance to $\mathcal{C}$, $\mathcal{J}$, or $\mathcal{F}$ without these additional controls.

Rubric Quality. Rubric quality asks whether $\mathcal{C}$ captures the intended objective before considering how a judge executes it. A controlled comparison should therefore hold $\mathcal{J}$ and $\mathcal{F}$ fixed wherever possible while varying the rubric. RUBRICBENCH supports this analysis through 1,147 difficult preference pairs paired with expert-authored atomic rubrics, enabling controlled comparison between self-generated and human-authored rubrics. It reports *Rubric Recall*, *Hallucination Rate*, and *Structural F1* to compare induced criteria with the references (Zhang et al. 2026b). These measures diagnose rubric–reference alignment, but do not establish validity by themselves: semantically equivalent rubrics may differ in formulation, and a discriminative rubric may still encode spurious requirements.

Execution Fidelity. Execution fidelity asks whether $\mathcal{J}$ applies a fixed rubric correctly and consistently. RUBRICEVAL directly targets this setting through rubric-level meta-

Table 5
Overview of benchmarks relevant to Rubric RM evaluation, compared by evaluation paradigm, domain, modality, and scale.

Benchmark	Evaluation	Domain	Modality	Scale
Component-level Meta-evaluation				
RubricBench (Zhang et al. 2026b)	*Pairwise*	*Multi-domain*	Language	1.1k
RUBRICEVAL (Pan et al. 2026)	*Pointwise*	*Multi-domain*	Language	3.5k
Rubric-based Benchmarks				
HealthBench (Arora et al. 2025)	*Pointwise*	*Medical*	Language	5k
PaperBench (Starace et al. 2025)	*Pointwise*	*Science*	Language	8.3k
ProfBench (Wang et al. 2025c)	*Pointwise*	*Professional*	Language	>7k
Auto-J (Li et al. 2024a)	*Pairwise*	*Multi-domain*	Language	1.4k
General Reward and Judge Benchmarks				
RewardBench (Lambert et al. 2025)	*Pairwise*	*Multi-domain*	Language	3.0k
MT-Bench (Zheng et al. 2023b)	*Pointwise & Pairwise*	*Multi-domain*	Language	80
RM-Bench (Liu et al. 2025b)	*Pairwise*	*Multi-domain*	Language	1.3k
JudgeBench (Tan et al. 2025)	*Pairwise*	*Multi-domain*	Language	350
M-RewardBench (Gureja et al. 2025)	*Pairwise*	*Multi-Lingual*	Language	2.9k
RAG-RewardBench (Jin et al. 2025b)	*Pairwise*	*RAG*	Language	1.5k
RewardMath (Kim et al. 2024b)	*BoN*	*MATH*	Language	483
RewardBench2 (Malik et al. 2025)	*Best-of-4*	*Multi-domain*	Language	1.9k
RMB (Zhou et al. 2024)	*Pairwise & BoN*	*Multi-domain*	Language	18.0k
PPE (Frick et al. 2025)	*Pairwise & BoN*	*Multi-domain*	Language	16.0k / 2.6k
VL-RewardBench (Li et al. 2025)	*Pairwise*	*Multi-domain*	Multi-modal	1.3k
Multimodal RewardBench (Michihiro, Luke, and Marjan 2025)	*Pairwise*	*Multi-domain*	Multi-modal	5.2k
MJ-Bench (Chen et al. 2025c)	*Pairwise*	*Image generation*	Multi-modal	2.4k

evaluation: given an instruction, a response, and one target rubric, a candidate judge predicts whether the response satisfies that rubric. The benchmark compares this binary prediction with quality-controlled reference labels and reports *Balanced Accuracy* and *Macro F1*, together with Easy/Hard splits and analyses of inter-judge variance (Pan et al. 2026). Because the rubric is fixed and human-authored or human-verified, these results primarily diagnose judge execution rather than rubric quality. Ordinary response-level preference accuracy does not provide the same isolation: a judge may compensate for a deficient rubric using internal knowledge, or misapply an otherwise valid rubric while still producing the correct final preference.

Aggregation Validity. Aggregation validity concerns whether $\mathcal{F}$ combines criterion-level judgments into an overall reward that is consistent with the intended evaluation objective. To examine aggregation separately from upstream components, the rubric and criterion-level judgments should be held fixed, and the aggregated result should be compared with an external reference, such as human overall judgments or predefined aggregation constraints. Existing general reward benchmarks usually provide only end-to-end labels and therefore conflate aggregation with rubric construction and judge execution (Lambert et al. 2025; Zheng et al. 2023b; Zhou et al. 2024; Malik et al. 2025); direct evaluation of aggregation remains limited.

Downstream Effectiveness. Downstream effectiveness asks whether rubric-mediated signals improve the artifact produced by any of the three pathways in Section 7: a training dataset, a learned policy, or an inference-time output. It contains two estimands that should not be conflated. *Reward or supervision utility* is measured by holding the down-

stream operator $\mathcal{O}$ fixed while varying the rubric-derived signal; *optimization effectiveness* proper is measured by holding the complete reward interface fixed while varying or diagnosing how $\mathcal{O}$ uses that signal. For data synthesis, models should be trained under matched recipes, source pools, data and compute budgets, and then assessed on downstream performance and data efficiency rather than on the apparent quality of synthesized examples alone. For policy learning, supervision utility should be evaluated under a fixed initial policy, optimization algorithm, rollout procedure, and data and compute budget, using an independent held-out evaluator. Comparisons should reflect the four design axes discussed in Section 7.2. In particular, process-sensitive evaluation should not be conflated with temporal credit assignment, since process-level judgments may still be aggregated into a single trajectory-level reward. Evaluating the optimizer itself instead requires holding the complete reward interface fixed while varying the optimization or credit-assignment mechanism. Relevant outcomes include final performance, sample efficiency, attribution fidelity, exploration, training stability, and reward–external-performance divergence across runs. For inference-time verification, gains from ranking, reranking, or iterative revision should be measured under matched candidate, tool-call, and compute budgets.

10. Challenges and Future Directions

10.1 *How can high-quality rubrics be identified under limited human annotation?*

Among the evaluation targets discussed in Section 9, rubric quality is particularly fundamental because it determines what the remaining reward-modeling pipeline is able to measure. Even if the judge applies every criterion consistently and the aggregation and optimization procedures operate as intended, an incomplete or misaligned rubric will still induce a reward that faithfully optimizes the wrong objective. Evaluating rubric quality is itself an open problem for *Rubric RMs*, especially under limited human supervision (Zhang et al. 2026b; Dhole and Agichtein 2026). In current practice, rubrics are often treated as intermediate content in reasoning, without explicit constraints or independent verification. This increases a typical *false-positive* risk: a model may reach the correct final judgment even when the rubric it relies on is incomplete, redundant, or otherwise unfaithful.

Prior work has explored various dimensions of rubric quality. These include discriminability (Zhang et al. 2025a); coverage, non-redundancy, and criteria trade-offs (Xie et al. 2025); correlation with the final answer (Sheng et al. 2026); validity (Sheng et al. 2026) and reliability (Li et al. 2026c); and value alignment (Zhang et al. 2026b; Wang et al. 2026a). However, there is still no unified definition of rubric quality, nor does a standardized evaluation framework exist. This lack of standardization makes it exceedingly difficult to evaluate rubric quality directly and equitably. Consequently, most studies are forced to assess overall rubric quality indirectly by observing the performance of rubric-guided judgments on downstream tasks (as discussed in §9). Developing a unified framework or designing meta-reward methods remains an important direction for future work.

Furthermore, systematic evaluations comparing different rubric construction and reasoning methodologies are still lacking, and foundational theoretical research (Mahmoud et al. 2026) regarding the underlying feasibility and mechanisms of rubrics remains extremely scarce. Ultimately, bridging these empirical and theoretical gaps is essential for transitioning Rubric RMs from heuristic tools into rigorous, trustworthy foundations for model alignment.

10.2 *How can common values be identified from human feedback and datasets?*

In many practical scenarios, *Rubric RMs* rely on fine-grained values to evaluate and improve individual instances. However, instance-specific rubrics are often too local to support robust reuse or transfer. While they may be effective for a particular example, they often fail to generalize across out-of-distribution tasks or datasets, limiting their ability to provide stable supervision at scale. This motivates the need to move beyond instance-level rubrics and identify more persistent dataset-level values across human feedback and data distributions. Recent efforts in this direction, such as the automatic construction of dataset-level rubrics by Xie et al. (2025) and Wang and Xiong (2025), suggest that common values may be extractable and reusable beyond individual cases. The broader reusability of dataset-level rubrics can reduce the need to generate a complete rubric for every individual instance, thereby lowering computational costs. However, this generality comes with a trade-off: dataset-level rubrics may obscure long-tail requirements and differences in values across user groups, and therefore cannot fully capture the personalized needs addressed by instance-level criteria. A promising direction is to combine reusable dataset-level rubrics, which establish a shared quality and safety floor, with instance-level criteria that adapt evaluation to local contexts and heterogeneous preferences.

Furthermore, evaluation will extend beyond the single-instance level to dataset analysis. Dataset-level extraction enables researchers to reverse-engineer the implicit human preference distribution from dataset-level rubrics, thereby allowing for a more scientific assessment of the quality and bias inherent in the dataset itself. Nevertheless, an important open question remains: how should such shared values be discovered, abstracted, and organized from human feedback and dataset-level supervision in a principled way? Additionally, existing research on dataset-level rubrics remains largely confined to open-ended question answering domains. Its effectiveness has not yet been validated in specialized vertical domains such as medicine or law. Consequently, successfully transferring common-value methodologies to these vertical domains represents another crucial open question.

10.3 *What is the appropriate structured representation of rubrics, and how can such structure be captured or learned effectively?*

Current research has not yet sufficiently addressed how rubrics should be represented as structured objects in complex multi-criteria evaluation. In practice, rubric criteria are rarely fully independent because it is difficult to construct a completely disentangled rubric. Nevertheless, many existing approaches still implicitly assume that each fine-grained criterion contributes independently to the final judgment. This suggests that an important research direction is to move beyond independent criterion formulations toward more structured representations of rubrics. Rather than treating rubrics as a simple collection of isolated dimensions, future work should explore how to represent and model the relational structure among criteria, including dependency, overlap, priority, and compositionality. Such structured representations are valuable not only for capturing non-linear criterion interactions, but also for improving generalization, interpretability, credit assignment, and context-sensitive evaluation. They can further provide an explicit basis for designing more faithful aggregation functions $\mathcal{F}$, allowing dependencies, priorities, overlaps, and veto relations among criteria to be reflected in the final reward rather than being reduced to an independent weighted sum.

10.4 *How can rubrics serve as adaptive scaffolds for policy learning?*

Current LLM reasoning faces a dual exploration dilemma: sparse solution spaces limit autonomous exploration in out-of-distribution tasks, while the lack of fine-grained guidance hinders capability acquisition in long-horizon reasoning. This suggests that the core challenge is not only exploration, but also guidance. Educational theories such as the Zone of Proximal Development (ZPD) (Vygotsky 1978) and scaffolding (Wood, Bruner, and Ross 1976) suggest that effective learning requires intermediate support, and rubrics are valuable precisely because they can make such intermediate goals explicit (Andrade 2005; Jonsson and Svingby 2007). From this perspective, *Rubric RMs* are better viewed not as static evaluators of response quality, but as structured capability scaffolds that support progressive policy improvement (Zhou et al. 2025; Li et al. 2026a; Fang et al. 2026; Bi et al. 2025; Zhang et al. 2026c; Yu et al. 2026a). Unlike traditional scalar rewards that often provide ambiguous signals during exploration, rubrics decompose response quality into independent, verifiable dimensions. This decoupled supervision acts as a "precision anchor," enabling the optimizer to explicitly penalize specific failures and advance directionally without degrading previously mastered skills. This shifts the focus from post hoc evaluation to joint optimization, where rubric-based rewards serve not only to assess outputs, but also to support capability acquisition during learning.

10.5 *What can traditional disciplines teach us about building and validating Rubric RMs?*

The preceding discussion illustrates how educational scaffolding can inform rubric-guided policy learning. More broadly, educational measurement, psychometrics, and instructional science offer design and validation principles that align with the four interfaces formalized in Section 3 and the four diagnostic targets introduced in Section 9. We organize these connections around rubric quality ($\mathcal{C}$), criterion-level execution ($\mathcal{J}$), aggregation validity ($\mathcal{F}$), and feedback and downstream effectiveness ($\mathcal{O}$).

10.5.1 Rubric Quality. Evidence-Centered Design suggests that assessment construction should begin by specifying what is to be evaluated, what observable evidence supports that evaluation, and what tasks can elicit the required evidence (Mislevy, Almond, and Lukas 2003). Applied to a *Rubric RM*, these components correspond to the intended quality represented by each criterion, the evidence in a response that indicates whether the quality is present, and the task conditions under which that evidence can be meaningfully observed. For example, for a factual accuracy criterion, the intended quality is the correctness of the response's factual claims, the relevant evidence is whether those claims agree with reliable references, and the task should require the response to make claims that can meaningfully be evaluated for factual accuracy. A criterion is poorly aligned with the task if it evaluates a quality that the response is not expected to demonstrate or requires evidence that is unavailable to the judge.

After the initial rubric is constructed, domain experts can evaluate whether each proposed criterion is essential to the intended evaluation target. Lawshe's Content Validity Ratio provides a formal way to summarize the proportion of experts who consider a criterion essential (Lawshe 1975). Low expert agreement indicates that the criterion should be removed or reconsidered. However, this analysis evaluates only the criteria that have already been proposed. It cannot by itself determine whether the rubric omits an important aspect of response quality, whether two criteria are unnecessarily

redundant, or whether their descriptions are sufficiently clear. These properties require additional expert review and empirical analysis of how the criteria operate in practice.

Rubric quality should not be treated as a property of the rubric text or its scores alone. What must be validated are the interpretations and uses based on the resulting scores (Kane 2013). For a *Rubric RM*, this requires examining the assumptions that connect the evidence observed in a response to its criterion-level scores and connect those scores to the resulting reward or decision. For example, if a safety criterion is used to reject unsafe responses or provide a training signal, researchers should test whether the criterion represents the intended safety requirement, whether the judge can recognize the relevant evidence, and whether the resulting score supports the intended use. A criterion may appear reasonable but still fail to support its intended use if its score is influenced by irrelevant response features or does not reliably distinguish different levels of the intended quality.

Combining these principles, an ideal rubric should be aligned with the intended human objective, express each relevant quality through evidence that can be observed in the response, and be appropriate for the task being evaluated. Its criteria should cover the important aspects of response quality, avoid irrelevant or unnecessarily overlapping requirements, and be sufficiently clear for consistent criterion-level judgment. Finally, the interpretations and uses of its scores should be supported by empirical evidence. A high-quality rubric therefore provides a defensible connection from the intended human objective, through observable response evidence and criterion-level scores, to the final reward or decision.

10.5.2 Criterion-Level Judgment. Research on scoring rubrics suggests that complex responses can be judged more consistently when evaluators use an analytic and task-specific rubric with clear performance descriptors and scored exemplars (Jonsson and Svingby 2007). For a *Rubric RM*, the central question at the judge level is whether each criterion is applied faithfully and consistently. A criterion-level score should depend on how well the response satisfies the criterion being evaluated, rather than on information that is irrelevant to that criterion. Therefore, two responses that satisfy the same criterion to a similar degree should receive similar scores, even if they differ in style, language, domain, or model source, provided that these differences are irrelevant to the criterion. Likewise, the same response should receive a stable score when the judge model or an otherwise equivalent evaluation setup changes. In practice, however, one judge model may be systematically stricter or more lenient than another, the judge's overall impression of a response may influence its scores on individual criteria, or responses that satisfy the same criterion to a similar degree may receive systematically different scores because of irrelevant differences in their presentation or source. Many-Facet Rasch Measurement, originally developed to study human raters, can be adapted to detect and quantify these patterns, including differences in judge severity, halo effects, and differential evaluation across response groups (Myford and Wolfe 2003). Generalizability Theory provides a complementary approach by estimating how much criterion-level score variation is associated with the judge model, evaluation prompt template, decoding run, response order, and their interactions (Brennan 1992). This analysis requires the same responses to be evaluated under systematically varied combinations of these conditions. It can reveal whether criterion-level judgments are stable and whether they are influenced by factors that should be irrelevant to the criterion. However, consistency alone does not establish correctness, so faithful criterion execution should also be evaluated against reliable criterion-level reference judgments.

10.5.3 Decision Aggregation. Criterion-referenced measurement provides a simple principle for aggregation: after the criterion-level scores are combined, the final reward should indicate whether a response meets predefined performance requirements, rather than only how it compares with the other responses in the current batch (Glaser 1963). To apply this principle, the aggregation rule should specify the required performance on each criterion and whether poor performance on one criterion can be offset by strong performance on others. For example, in a *Rubric RM*, a serious safety failure should not be offset by good writing style, whereas a minor weakness in wording may be offset by strong factual accuracy and completeness. A rule that requires every essential criterion to meet its requirement is called conjunctive, while a rule that allows strengths on some criteria to compensate for weaknesses on others is called compensatory (Haladyna and Hess 1999). When reliable reference decisions are available, the aggregation rule can be evaluated by checking how often it incorrectly accepts or rejects a response and whether it produces consistent decisions under repeated evaluations (Livingston and Lewis 1995).

10.5.4 Feedback and Optimization. Formative assessment distinguishes between identifying a performance gap and providing feedback that can actually reduce it. Effective feedback requires a clear target, information about the gap between the current response and that target, and an action that can help close the gap (Sadler 1989). Applied to a *Rubric RM*, the rubric defines the desired performance, criterion-level judgments identify which requirements remain unmet, and the resulting diagnosis should determine the next optimization action. For example, failure on a factual support criterion may trigger evidence retrieval or revision of unsupported claims, while a failure on instruction following may require the response to be regenerated under the missing constraint. At the training level, the same diagnosis may be used to select relevant training examples or assign a more targeted learning signal. An unmet criterion is therefore useful not merely because it describes an error, but because it identifies what should be changed next.

Scaffolding provides a complementary principle for deciding how much guidance to provide. It suggests that support should focus on the parts of a task that exceed the learner's current capability, while leaving the learner to handle parts that are already within reach (Wood, Bruner, and Ross 1976). For a *Rubric RM*, criterion-level diagnoses can similarly direct guidance, computation, or learning signals toward the specific requirements that the policy cannot yet satisfy. This avoids repeatedly intervening in criteria that the policy already handles reliably and allows the optimization process to concentrate on its current bottlenecks.

Feedback, however, does not necessarily improve performance. Research on feedback interventions shows that feedback can reduce performance when it directs attention away from the task or toward information that does not help correct the underlying problem (Kluger and DeNisi 1996). For a *Rubric RM*, an immediate increase in rubric-based reward may therefore reflect a superficial response to the feedback or adaptation to evaluator-specific patterns rather than a genuine improvement in capability. Rubric-guided optimization should consequently be evaluated by whether it improves performance on subsequent and held-out tasks, whether the improvement remains under independent evaluation, and whether addressing one criterion causes regressions on criteria that were previously satisfied. An ideal feedback and optimization mechanism should turn criterion-level diagnoses into actionable interventions, concentrate support on current capability gaps, and demonstrate improvements that extend beyond the immediate reward signal.

11. Conclusion

This paper has presented a comprehensive survey of *Rubric RMs* from four core perspectives: foundations, modeling, reasoning, and optimization. By treating the rubric as an explicit intermediate representation, this organization separates rubric quality, criterion-level execution, aggregation, and policy optimization, preventing final performance from obscuring errors in individual components. We have also summarized the datasets and benchmarks used to develop and evaluate Rubric RMs. Drawing on established principles from education and psychometrics, we identified key challenges involving rubric validity, judge reliability, structured aggregation, and adaptive scaffolding. Addressing these challenges will be important for the continued development of Rubric RMs as interpretable and reliable mechanisms for guiding reasoning, exploration, and learning in complex, open-ended domains.

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